

CLOUD & DEVSECOPS

CAPSTONE

Project 1 - Cloud Readiness & Migration Assessment

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Classification: Confidential Enterprise Document

Duration: 1-Week Intensive Infrastructure Discovery

1.0 Deliverable 1: Cloud Assessment Report

1.1 Executive Summary

This Cloud Assessment Report establishes the foundational baseline required to execute a highly secure, enterprise-scale public cloud migration. Conducted by Lukman Ali over an intensive one-week discovery period, this evaluation examines the organization's current physical and virtualized computing environments to assess readiness for Microsoft Azure or Amazon Web Services (AWS) adoption.

This document serves as the formal technical sign-off required before advancing into migration execution. It meticulously details existing architectural limitations, scores enterprise operational maturity, projects total cloud migration financial modeling (FinOps), and proposes a heavily fortified enterprise landing zone.

TABLE 1.1: ASSESSMENT KEY FINDINGS SUMMARY

Information: High-level executive overview of critical assessment metrics and strategic migration priorities identified during the initial discovery phase.

KEY ASSESSMENT FINDING	CURRENT STATUS / TARGET VALUE	BUSINESS PRIORITY
Overall Cloud Maturity	3.2 / 5.0 — Intermediate Rating	High
Applications Cloud-Ready	6 of 12 critical business applications	High
Security Gaps Identified	MFA, IAM, and Core Encryption gaps	Critical
Estimated Cost Reduction	28% to 44% annually post-migration	High
Legacy OS / EOL Software	8 servers running end-of-life operating systems	Critical
Target Migration Timeline	18 months (Executed securely across 3 phases)	Medium

1.2 Core Business Objectives & Drivers

The imperative for this transformation is driven by acute hardware obsolescence and restrictive capital expenditure (CapEx) models. By transitioning to a dynamic, scalable operational expenditure (OpEx) architecture, the organization intends to radically improve application resilience, completely eliminate rigid physical data centers, and embed rigorous DevSecOps methodologies into daily engineering practices within an aggressive 18-month timeline.

Additionally, regulatory pressures demand immediate action. Failing on-premises infrastructure is currently jeopardizing the organization's adherence to stringent data security frameworks, particularly PCI-DSS and GDPR. Transitioning to the cloud unlocks advanced automated compliance controls, ensuring sensitive corporate data is guarded by zero-trust architecture.

TABLE 1.2: EXECUTIVE MIGRATION OBJECTIVES

Information: Core strategic drivers guiding the enterprise transition from on-premises to cloud.

STRATEGIC BUSINESS DRIVER	TARGET CLOUD OUTCOME	ASSESSED PRIORITY
Financial Agility (OpEx)	Transition IT spend from CapEx to a flexible, consumption-based OpEx model.	High
Infrastructure Resilience	Eradicate single points of failure utilizing multi-zone cloud architecture.	Critical
Security Modernization	Implement Zero-Trust networks, MFA, and automated compliance auditing.	Critical
Operational Automation	Enable Infrastructure as Code (IaC) and comprehensive CI/CD pipelines.	High

1.3 Current State Architectural Baseline

The existing on-premises data center operates as a traditional, hybrid physical-virtual infrastructure. Core computing resources are distributed across aging bare-metal Dell and HP servers, while roughly 83% of the workload environment is managed utilizing VMware vSphere 6.7 hypervisors. This environment currently exhibits severe operational friction.

A critical architectural bottleneck is the continued reliance on End-of-Life (EOL) operating systems. The discovery phase identified eight foundational servers stranded on Windows Server 2008 R2. Without vendor security updates, these nodes represent an immense cyber liability. Furthermore, disaster recovery protocols rely exclusively on tape-based media, forcing an unacceptable 4-hour Recovery Time Objective (RTO).

TABLE 1.3: CURRENT ENVIRONMENT ARCHITECTURE

Information: A structural mapping of the existing physical and virtualized data center parameters.

INFRASTRUCTURE DOMAIN	CURRENT DATA CENTER BASELINE	IDENTIFIED BOTTLENECK
Compute Hardware	24 Physical Servers (Dell/HP)	High maintenance; aging warranties.
Virtualization Tier	68 VMs via VMware vSphere 6.7	Hypervisor licensing costs escalating.
Operating Systems	Windows Server / RHEL 7 & 8	8 EOL Windows 2008 R2 servers present.
Enterprise Storage	180 TB combined SAN and NAS	Costly to expand; high physical footprint.
Disaster Recovery	Tape backup (4-hour RTO)	No automated off-site failover capability.

1.4 Infrastructure & Hardware Inventory

A granular analysis of the hardware inventory strongly dictates the required migration methodology. The high rate of VMware virtualization (accounting for 20 out of 24 core servers) structurally prepares the vast majority of the web and application tiers for rapid, block-level 'Lift and Shift' rehosting into cloud-native virtual machines.

Conversely, the networking perimeter requires a complete redesign before workloads can be safely migrated. The legacy Cisco ASA edge firewalls currently enforce 38 overpermissive inbound routing rules that violate modern Zero-Trust architectures. These must be purged and replaced with strict Cloud Network Security Groups (NSGs) tied to application-specific ports.

TABLE 1.4: SERVER & NETWORK READINESS MATRIX

Information: Categorical breakdown evaluating the cloud migration eligibility of core infrastructure.

RESOURCE CATEGORY	INVENTORY VOLUME	CLOUD ELIGIBILITY & ARCHITECTURAL STRATEGY
Web / App Servers	8 Servers (100% Virtualized)	High Eligibility: Rehost to scalable Cloud VMs.
Database Servers	6 Servers (Mixed status)	Low Eligibility: Oracle 11g mandates complex replatforming.
Domain Controllers	3 Servers (Virtualized)	Medium Eligibility: Requires Entra ID hybrid sync setup.
Network Bandwidth	500 Mbps Dedicated Line	Adequate: Sufficient for initial phased data replication.
Edge Firewalls	Legacy Cisco ASA	Inadequate: 38 faulty rules require total zero-trust rebuild.

1.5 Application Portfolio Assessment

The corporate application portfolio consists of twelve primary workloads, each carrying a unique risk profile and architectural dependency. A meticulous application rationalization strategy was executed to prevent the dangerous migration of poorly optimized legacy code, ensuring that cloud-native capabilities are maximized.

The assessment dictates that low-risk systems, such as the internal CRM and File Servers, are prime candidates for immediate rehosting. However, the sprawling Enterprise Resource Planning (ERP) platform is a deeply monolithic application coupled to an aging Oracle database. This system requires high-risk, intensive refactoring to decouple its services into scalable containers.

TABLE 1.5: WORKLOAD RATIONALIZATION PORTFOLIO

Information: Strategic mapping of business applications to their target cloud transformation patterns.

TARGET APPLICATION	WORKLOAD TYPE	DATABASE ENGINE	ASSIGNED CLOUD STRATEGY
CRM System	Web Application	MSSQL 2019	Rehost (Lift & Shift) — Low Risk
Core ERP Platform	Monolithic App	Oracle 11g	Refactor (Microservices) — High Risk
HR Staff Portal	Web Frontend	MySQL 8.0	Replatform (Managed PaaS) — Medium Risk
Analytics / BI	Reporting Tool	MSSQL Data Whse	Replatform (SaaS/PaaS) — Low Risk
Legacy Billing	Desktop Client	Oracle 11g	Retire (Decommission) — High Risk
Payment Module	Secure Gateway	MSSQL 2019	Refactor (PCI-DSS Scoped) — Critical Risk

1.6 Cybersecurity & Compliance Review

The cybersecurity assessment uncovered severe vulnerabilities that act as hard blockers for cloud migration. The most critical failure point is the complete absence of Multi-Factor Authentication (MFA) across all internal administrative accounts. In a decentralized cloud environment, relying exclusively on password authentication guarantees a breach.

Data protection mechanisms are similarly deficient. Four primary database servers currently operate without Transparent Data Encryption (TDE), violating core enterprise data protection standards. These gaps directly impact regulatory compliance; the organization currently fails Payment Card Industry (PCI-DSS) standards due to inadequate tokenization and encryption controls.

TABLE 1.6: CRITICAL SECURITY VULNERABILITIES

Information: Identification of severe cyber vulnerabilities and the required cloud remediation pathways.

SECURITY VULNERABILITY	CURRENT AUDIT STATUS	MANDATORY CLOUD REMEDIATION
Identity Access (MFA)	No MFA enabled on 100% of admin logins.	Enforce MFA globally via an Identity Provider (IdP).
Data Encryption	4 major database servers unencrypted at rest.	Enable AES-256 Transparent Data Encryption (TDE).
Network Routing Rules	38 overpermissive inbound firewall rules.	Purge rules; institute strict Zero-Trust cloud network models.
Admin Account Sprawl	Excessive administrative users (47 total).	Implement Role-Based Access Control (RBAC) and PAM.
PCI-DSS Compliance	Failing due to missing tokenized data controls.	Refactor application to isolate and encrypt payment data.

1.7 Dependency Mapping & Integrations

Cloud migrations frequently fail when administrators misunderstand complex application interoperability. Workloads cannot be transported arbitrarily across the hybrid boundary; they must be migrated in carefully orchestrated "move groups" to prevent induced cloud latency from breaking critical internal API connections.

For example, the Payment Module relies on high-speed, secure API handshakes with the CRM system. Separating these two systems across a WAN link during a phased migration would result in unacceptable transaction latency. Therefore, tightly coupled systems must be grouped, replicated, and cut over synchronously into the target Cloud Virtual Network.

TABLE 1.7: APPLICATION INTEROPERABILITY MAP

Information: Documentation of deep system relationships to dictate safe, grouped workload migrations.

TARGET APPLICATION	COUPLED DEPENDENCY	INTEGRATION PROTOCOL	MIGRATION EXECUTION STRATEGY
CRM System	MSSQL DB, AD Auth	REST API calls	Migrate alongside primary Domain Controllers.
ERP Platform	Oracle DB Database	Direct DB queries	Requires synchronized cutover with database tier.
HR Portal	MySQL DB, AD Auth	LDAP / SAML Sync	Migrate in later phases once Identity sync is stable.
Payment Module	CRM System Data	Internal Secure API	Strict requirement to peer within the same secure VNet.

2.0 Deliverable 2: Maturity Scoring Sheet

2.1 Enterprise Evaluation Framework

To accurately benchmark the organization's readiness to operate a complex cloud ecosystem, a formal Cloud Maturity Assessment was executed. This framework evaluates enterprise capabilities across six fundamental operating dimensions: Strategy & Governance, People & Skills, Process & Operations, Technology, Security, and Financial Management.

Each dimension is quantified on a strict 1 to 5 scale. A score of 1 represents a highly reactive, unmanaged state heavily reliant on manual human intervention. Conversely, a score of 5 indicates a fully optimized, automated environment driven by Continuous Integration/Continuous Deployment (CI/CD) pipelines and cloud-native practices.

TABLE 2.1: MATURITY SCALE GRADING MATRIX

Information: The standardized rubric utilized to objectively quantify operational and technical readiness.

SCORE	MATURITY LEVEL	DETAILED OPERATIONAL CHARACTERISTICS
1	Ad-Hoc	No formal processes. Operations are highly reactive and completely undocumented.
2	Developing	Basic processes exist but remain disjointed, siloed, and heavily manual.
3	Defined	Processes are documented and repeatable. Foundational automation exists.
4	Managed	Operations are monitored via telemetry and consistently applied across teams.
5	Optimized	Full automation, robust IaC deployment, and proactive continuous improvement.

2.2 Strategy, People & Process Dimensions

The cultural and organizational dimensions yielded critical vulnerabilities. The "People & Skills" category scored an alarmingly low 2.2 out of 5.0. Out of a twelve-person infrastructure team, only one engineer possesses a relevant cloud certification. Transitioning to DevSecOps requires an immediate, massive investment in staff training and change management.

Furthermore, the absence of a board-approved Cloud Strategy severely impacts governance. Without centralized tagging standards, defined deployment policies, and a dedicated Cloud Center of Excellence (CCoE) to act as the architectural gatekeeper, the migration risks resulting in decentralized shadow IT and unmanageable cloud sprawl.

TABLE 2.2: GOVERNANCE & OPERATIONAL SCORING

Information: Evaluation of strategic planning, engineering skillset availability, and workflow maturity.

ASSESSMENT DIMENSION	SUB-CRITERIA ANALYZED	ASSESSED SCORE & EVIDENCE
Strategy & Gov.	Cloud strategy documentation	2.6 / 5.0 — No formal enterprise strategy exists.
Strategy & Gov.	Resource tagging & policies	2.6 / 5.0 — Zero guardrails established for cloud use.
People & Skills	Certified cloud engineers	2.2 / 5.0 — Only 1 AWS certification across 12 staff.
People & Skills	Cloud Center of Excellence	2.2 / 5.0 — No CCoE committee established.
Process & Ops.	CI/CD pipeline maturity	2.6 / 5.0 — Heavily reliant on manual code deployments.

2.3 Technology, Security & FinOps Dimensions

The Technology dimension scored reasonably well (3.4) due to deep historical investments in VMware virtualization, providing a solid launchpad for IaaS migrations. However, modern deployment mechanics are absent. The total lack of Infrastructure as Code (IaC) utilities such as Terraform prevents automated, reproducible, and state-locked infrastructure provisioning.

Financial management currently focuses purely on hardware capital depreciation. The organization lacks the modern FinOps paradigms required to control elastic cloud billing. Without automated anomaly detection and budget enforcement alerts, the enterprise is highly susceptible to severe, unexpected cloud billing spikes post-migration.

TABLE 2.3: TECHNOLOGY, SECURITY & FINANCE SCORING

Information: Assessment of structural IT architecture, cybersecurity postures, and cost modeling maturity.

ASSESSMENT DIMENSION	SUB-CRITERIA ANALYZED	ASSESSED SCORE & EVIDENCE
Technology	Virtualization readiness	3.4 / 5.0 — 83% virtualization provides a strong baseline.
Technology	Infrastructure as Code	3.4 / 5.0 — Zero usage of Terraform, ARM, or Bicep.
Security	Identity access controls	2.6 / 5.0 — AD exists, but absolutely no MFA implemented.
Security	Security logging (SIEM)	2.6 / 5.0 — No centralized log aggregation platform deployed.
Finance / FinOps	Automated cloud budgeting	3.2 / 5.0 — Excellent CapEx tracking, but zero OpEx controls.

2.4 Aggregated Assessment Scoring Summary

Synthesizing all six operational dimensions results in an aggregated Enterprise Cloud Maturity Score of 2.7 out of 5.0. This classifies the organization squarely within the "Developing" or "Intermediate" tier. The data indicates a stark divergence between technical feasibility and organizational readiness.

While the highly virtualized server infrastructure is technically prepared to be copied into cloud storage blobs, the engineering teams are culturally and procedurally unprepared to operate those systems securely once migrated. Attempting to execute this migration without bridging these operational gaps guarantees massive security and financial failures.

TABLE 2.4: OVERALL ENTERPRISE MATURITY CLASSIFICATION

Information: The consolidated final assessment scores determining the organization's baseline cloud readiness tier.

EVALUATION DIMENSION	CALCULATED SCORE	PERCENTAGE RESULT	TIER CLASSIFICATION
Financial Management	16 out of 25	64%	Defined State
Technology & Architecture	14 out of 25	56%	Defined State
Strategy & Governance	13 out of 25	52%	Developing State
Process & Operations	13 out of 25	52%	Developing State
Security & Compliance	13 out of 25	52%	Developing State
People & Technical Skills	11 out of 25	44%	Developing State
AGGREGATED OVERALL RATING	80 out of 150	53%	2.7 / 5.0 (Intermediate)

2.5 Critical Gap Analysis & Prioritization

The Gap Analysis bridges the current assessed state (2.7 rating) to the targeted mature state (4.0+ rating) required to sustain a secure enterprise cloud architecture. This roadmap prioritizes remediation tasks based strictly on immediate security risk and operational blocking factors.

Resolving the 'People & Skills' deficit through mandatory cloud certification curriculums is as imperative as the technical deployments. Furthermore, replacing error-prone manual administrative routines with automated CI/CD deployment pipelines (utilizing Azure DevOps or GitHub Actions) is a non-negotiable prerequisite prior to Phase 2 of the migration.

TABLE 2.5: PRIORITIZED REMEDIATION ACTION PLAN

Information: Mandatory strategic remediation tasks mapped against required maturity milestones.

STRATEGIC GAP AREA	TARGET SCORE	EXECUTION PRIORITY	MANDATORY ACTION REQUIRED
Cloud Governance	4.0	Critical	Draft, finalize, and secure board approval for a multi-year cloud strategy.
Security Defenses	4.5	Critical	Execute an immediate global MFA and PAM rollout across all IT admins.
Engineering Skills	4.0	High	Launch aggressive, mandatory cloud certification training (e.g., AZ-900).
Deployment Automation	4.0	High	Build robust CI/CD pipelines to replace all manual server configurations.
FinOps Architecture	4.0	Medium	Implement Azure Cost Management alerts to prevent budget overruns.

3.0 Deliverable 3: TCO / ROI Comparison

3.1 Financial Migration Assumptions

A rigorous Total Cost of Ownership (TCO) analysis fundamentally justifies the commercial viability of a large-scale public cloud transformation. This detailed financial modeling contrasts the immense capital burden of refreshing the on-premises data center against the dynamically optimized, consumption-based costs of an enterprise cloud environment.

The accuracy of these projections hinges on executing strict FinOps methodologies. The core assumption driving these savings is "Right-Sizing." Historically, on-premises physical servers are drastically over-provisioned. By precisely matching cloud compute SKUs to the actual historical 45% CPU utilization metric, massive computational waste is instantly eradicated.

TABLE 3.1: FUNDAMENTAL FINOPS ASSUMPTIONS

Information: Core architectural parameters and commercial discount strategies utilized to accurately forecast cloud expenditure.

FINANCIAL ASSUMPTION PARAMETER	EXECUTION STRATEGY & COST CONTROL METHOD
Compute Instance Right-Sizing	Cloud VMs sized strictly based on telemetry to eliminate physical waste.
Reserved Instance Discounts	3-year Reserved Instance (RI) commitments applied for 40% compute discounts.
Enterprise License Portability	Aggressive use of Bring Your Own License (BYOL) via Azure Hybrid Benefit.
Corporate Application SaaS Shift	Replacing costly local Exchange Servers entirely with M365 SaaS licensing.
Network Egress Calculation	Financial model incorporates a conservative 2 TB/month data outbound fee.

3.2 Target Annual Cloud Operating Costs

A comprehensive line-item audit reveals profound financial inefficiencies within the legacy infrastructure. Sustaining the physical data center requires continuous, immense capital depreciation. Annual hardware lifecycle replacements cost \$180,000, while physical facility overhead (commercial power, HVAC cooling systems, physical security) drains an additional \$65,000 annually without providing any competitive advantage.

The proposed target cloud architecture completely eliminates these physical burdens. Furthermore, applying the Bring Your Own License (BYOL) protocol slashes Microsoft Windows and SQL database licensing fees by over \$44,000 per year. Despite introducing new operational cost vectors—such as elastic compute billing and storage egress—the overall IT run-rate drops by a highly impactful 44%.

TABLE 3.2: ANNUAL OPERATING COST BREAKDOWN (ON-PREMISES VS. CLOUD)

Information: Direct line-item comparison of current physical expenditure versus projected cloud optimization.

EXPENDITURE CATEGORY	ON-PREMISES (ANNUAL)	CLOUD ESTIMATE (ANNUAL)	CALCULATED NET IMPACT
Hardware Refresh & Facilities	\$245,000	\$0	\$245,000 CapEx Eliminated
VMware Hypervisor Licensing	\$42,000	\$0	\$42,000 Saved via Native IaaS
Windows & SQL Server Licenses	\$63,000	\$18,900	\$44,100 Saved via Hybrid Benefit
Legacy Oracle Licensing	\$48,000	\$0	\$48,000 Saved by Replatforming
Storage Arrays & Tape Backup	\$50,000	\$24,000	\$26,000 Saved via Cloud Vaults

3.3 Three-Year FinOps ROI Projection

Evaluating this transformation across a multi-year timeline reveals the compounding financial dominance of the cloud architecture. Year 1 incurs elevated operational costs (\$550,000) due to the mandatory requirement of dual-running; the organization must power the legacy hardware while concurrently paying for the new cloud infrastructure during migration data syncs.

However, by Year 3, the organization cleanly sidesteps a massive, unavoidable \$780,000 legacy hardware refresh cycle, driving the total 3-year cumulative savings to \$906,100. Ensuring these projections remain accurate requires establishing rigid, automated FinOps controls, such as auto-scaling development clusters down to zero capacity during non-business hours.

TABLE 3.3: MULTI-YEAR TOTAL COST OF OWNERSHIP PROJECTION

Information: Three-year cumulative forecast highlighting the massive financial return generated by eliminating hardware refreshes.

FINANCIAL YEAR PHASE / MILESTONE	ON-PREMISES RUN-RATE	TARGET CLOUD RUN-RATE	ANNUAL NET SAVING
Year 1 (Dual-running migration phase)	\$740,000	\$550,000	\$190,000 Saved
Year 2 (Steady-state optimization)	\$740,000	\$413,900	\$326,100 Saved
Year 3 (Hardware refresh completely avoided)	\$780,000	\$390,000	\$390,000 Saved
3-YEAR AGGREGATED TOTAL	\$2,260,000	\$1,353,900	\$906,100 (40% Drop)
Projected Financial Break- Even	—	Month 11	Aggressive ROI target
Total 3-Year Net ROI Percentage	—	853% Return	Based on initial \$95k CapEx

4.0 Deliverable 4: Migration & Remediation Plan

4.1 Architectural Cloud Strategy (The 6 R's)

To safely execute a migration of this magnitude without disrupting corporate operations, a disciplined framework must be applied to every unique workload. This roadmap utilizes the enterprise "6 R's" methodology to dictate the specific architectural treatment of each underlying system.

Low-risk, highly compatible systems such as the CRM platform are slated for "Rehosting" to accelerate the migration velocity. Conversely, high-value monolithic systems like the ERP platform require intensive "Refactoring", breaking the monolithic codebase down into dynamic, containerized microservices orchestrated by Kubernetes to maximize scalability and fault isolation.

TABLE 4.1: STRATEGIC "6 R'S" APPLICATION MAPPING

Information: Categorical assignment of strict architectural transformation patterns to existing business workloads.

ARCHITECTURAL STRATEGY	DEFINITION & EXECUTION APPROACH	ASSIGNED APPLICATION TARGETS
Rehost (Lift & Shift)	Replicate server exactly as-is to cloud IaaS Virtual Machines.	CRM System, Main File Servers
Replatform	Execute minor configuration changes to utilize managed PaaS.	HR Portal, BI Analytics Platform
Refactor / Rearchitect	Completely redesign code to utilize scalable cloud microservices.	ERP Platform, Payment Modules
Repurchase	Abandon legacy software and adopt modern SaaS equivalents.	Exchange Email (Moving to M365)
Retire & Terminate	Safely decommission platforms providing zero active business value.	Legacy Billing Application App

4.2 Enterprise Landing Zone Deployment

Prior to migrating a single byte of sensitive production data, an enterprise-grade Cloud Landing Zone must be engineered. This specialized environment serves as the highly secure, scalable foundation for all future cloud deployments, baking essential compliance, networking, and identity controls directly into the foundational infrastructure code.

The proposed architecture relies heavily on a Hub-and-Spoke Virtual Network topology. The centralized Hub acts as the absolute ingress/egress transit point, housing the Azure Firewall and VPN Gateways. Production applications will reside in isolated Spoke networks, fiercely protected by localized Network Security Groups enforcing a strict Zero-Trust traffic policy.

TABLE 4.2: FOUNDATIONAL LANDING ZONE ARCHITECTURE

Information: Structural design mandates dictating the configuration of the secure cloud perimeter environment.

LANDING ZONE COMPONENT	STRATEGIC IMPLEMENTATION MANDATE	PRIMARY CLOUD SERVICE UTILIZED
Identity & Authentication	Hybrid AD configuration synchronizing local user identities.	Azure Active Directory (Entra ID)
Network Topology	Hub-and-Spoke VNet layout enforcing traffic segmentation.	Azure VNet + Core VPN Gateway
Security Baseline	Absolute zero-trust packet inspection and vulnerability scanning.	Microsoft Defender for Cloud
Governance & Costing	Hierarchical tags preventing rogue deployments and tracking spend.	Azure Policy & Management Groups
Backup & Resilience	Automated geo-redundant vaults targeting a 1-hour Recovery Point.	Azure Backup + Site Recovery

4.3 18-Month Phased Migration Roadmap

The transformation is sequentially divided into three distinct execution phases spanning 18 months, meticulously engineered to minimize operational disruption. Phase 1 centers on establishing the Landing Zone and executing "Quick Wins" to build institutional momentum, such as migrating the File Servers and immediately enforcing global MFA policies.

Phase 2 systematically replatforms the mid-tier databases and formalizes the CI/CD deployment pipelines. Phase 3, the most complex stage, executes the intensive refactoring of the monolithic ERP system, secures definitive PCI-DSS compliance, and culminates in the total physical decommissioning of the legacy data center facility.

TABLE 4.3: COMPREHENSIVE 18-MONTH EXECUTION ROADMAP

Information: The sequential chronological timeline orchestrating workload migrations and critical security upgrades.

EXECUTION PHASE & MILESTONE	ASSIGNED TEAM OWNER	TARGET DURATION
Phase 1: Foundation (Months 1–4)		
- Engineer secure cloud Landing Zone & governance	Cloud Architect	3 Weeks
- Execute mandatory MFA rollout across all accounts	Security Lead	Critical (Month 1)
- Migrate Corporate File Server natively to Azure Files	Infra Engineer	2 Weeks
Phase 2: Core Migration (Months 5–10)		
- Replatform HR Portal to scalable Azure App Service	Dev Team	6 Weeks
- Establish robust CI/CD code deployment pipelines	DevOps Engineer	4 Weeks

5.0 Strategic Recommendations & Conclusion

Based on the exhaustive findings derived from this intensive one-week infrastructure assessment, the fundamental conclusion is indisputable: maintaining the organization's current operational trajectory within the legacy on-premises data center presents unacceptable business, security, and financial liabilities. Continuing to operate mission-critical business applications on End-of-Life (EOL) operating systems, compounded by severe security deficiencies such as the total absence of Multi-Factor Authentication (MFA) and unencrypted storage volumes, leaves the enterprise profoundly and continuously vulnerable to catastrophic cyber breaches and prolonged operational paralysis.

While the organization's current Cloud Maturity Score sits at a modest 2.7 out of 5.0 (Intermediate), the physical technical foundation—driven by an 83% virtualization rate—is entirely capable of sustaining a phased, highly secure cloud transformation. The critical gaps identified in advanced engineering skills and automated DevOps processes can, and must, be aggressively remediated in parallel with early Phase 1 deployment efforts. This will be achieved through heavily mandated training curriculums and the immediate establishment of a powerful Cloud Center of Excellence (CCoE) to govern architectural standards.

Furthermore, the financial justification for this massive modernization effort is irrefutable. By rapidly transitioning the enterprise away from a highly inefficient, rigid Capital Expenditure (CapEx) hardware life-cycle model and embracing the elasticity of an Operational Expenditure (OpEx) cloud architecture, the business is empirically projected to realize an immediate 44% reduction in annual IT operating costs. Baseline annual expenditure will radically drop from \$740,000 to an optimized \$413,900. Over a comprehensive 3-year strategic horizon, by aggressively leveraging architectural Right-Sizing, Azure Hybrid Benefit (BYOL) licensing protocols, and long-term Reserved Instances, the organization will secure a staggering \$906,100 in cumulative net savings, breaking even on its initial capital migration investment in an unprecedented 11 months.

Final Enterprise Recommendation: It is emphatically recommended that executive leadership immediately approve the execution of Phase 1 of the 18-month Migration Roadmap. The immediate and uncompromising priority must be the construction of a hardened, Zero-Trust